

1) Assuming the flow is from the bottom to the top.

Applying the energy equation:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + \cancel{z_1} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 - h_L$$

$$\frac{\cancel{\rho g h_1}}{\cancel{\rho g}} = \frac{\cancel{\rho g h_2}}{\cancel{\rho g}} + z_2 - h_L$$

$$3 = 1 + 1.5 - h_L$$

$$h_L = 0.5$$

$$2) h_L = \frac{1.2v^2}{2g}$$

$$h_p = 20 - 2000Q^2$$

$$\cancel{\frac{p_1}{\rho g}} + \cancel{\frac{v_1^2}{2g}} + z_1 + h_p = \cancel{\frac{p_2}{\rho g}} + \frac{v_2^2}{2g} + \cancel{z_2} + h_L$$

$$6 + h_p = \frac{v_2^2}{2g} + h_L$$

$$6 + 20 - 2000Q^2 = \frac{v^2}{2g} + \frac{1.2v^2}{2g}$$

$$26 - 2000Q^2 = \frac{2.2v^2}{2g}$$

$$g(26 - 2000Q^2) = 1.1v^2$$

$$26 - 2000Q^2 = \frac{1.1\left(\frac{Q}{A}\right)^2}{g}$$

$$26 = \frac{1.1Q^2}{A^2g} + 2000Q^2$$

$$\frac{1.1Q^2}{\left(\left(\frac{0.07}{2}\right)^2\pi\right)^2g} + 2000Q^2 = 26$$

$$Q = 0.05212050388 \\ \approx 0.052 \text{ m}^3 \text{ s}^{-1}$$

$$3) \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + \cancel{z_2} + h_{\text{turbine}} - h_L$$

$$P_2 = \rho g h = 13.6 \times 10^3 \times 9.81 \times \underset{\substack{\uparrow \\ \text{vacuum}}}{-25 \times 10^{-2}} \\ = -33354 \text{ Pa}$$

$$V_1 = \frac{Q}{A_1} = \frac{4.5}{\left(\frac{1}{2}\right)^2 \pi} \\ = 5.729577951 \text{ ms}^{-1}$$

$$V_2 = \frac{Q}{A_2} = \frac{4.5}{\left(\frac{1.2}{2}\right)^2 \pi} \\ = 3.978873577 \text{ ms}^{-1}$$

$$h_{\text{turbine}} = \frac{W}{\rho g Q} = \frac{1.9 \times 10^6}{1000 \times 9.81 \times 4.5} \\ = 43.03998188 \text{ m}$$

$$\frac{415 \times 10^3}{1000 \times 9.81} + \frac{5.73^2}{2 \times 9.81} + 3 = \frac{-33354}{1000 \times 9.81} + \frac{3.98^2}{2 \times 9.81} + 43.04 - h_L$$

$$h_L = 6.53008074 \\ \approx 6.53 \text{ m}$$

$$\therefore \text{Power loss} = \rho g Q h_L$$

$$= 1000 \times 9.81 \times 4.5 \times 6.53 \\ = 288.2704143 \text{ kW} \\ \approx 288.3 \text{ kW}$$

$$4) \dot{m}_3 \left(\frac{P_3}{\rho} + \frac{V_3^2}{2} + \cancel{z_3 g} \right) = \dot{m}_1 \left(\frac{P_1}{\rho} + \frac{V_1^2}{2} + \cancel{z_1 g} \right) + \dot{m}_2 \left(\frac{P_2}{\rho} + \frac{V_2^2}{2} + \cancel{z_2 g} \right) - \dot{W}_L$$

$$\dot{m}_3 \left(\frac{P_3}{\rho} + \frac{V_3^2}{2} \right) = \dot{m}_1 \left(\frac{P_1}{\rho} + \frac{V_1^2}{2} \right) + \dot{m}_2 \left(\frac{P_2}{\rho} + \frac{V_2^2}{2} \right) - \dot{W}_L$$

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\rho V_3 A_3 = \rho V_1 A_1 + \rho V_2 A_2$$

$$2(0.6) = 1.4(0.5) + 0.2 V_2$$

$$V_2 = 2.5 \text{ m/s}$$

$$\begin{aligned} \dot{m}_3 &= 1000 \times 2 \times 0.6 \\ &= 1200 \end{aligned}$$

$$\begin{aligned} \dot{m}_1 &= 1000 \times 0.5 \times 1.4 \\ &= 700 \end{aligned}$$

$$\begin{aligned} \dot{m}_2 &= 1000 \times 0.2 \times 2.5 \\ &= 500 \end{aligned}$$

$$1200 \left(\frac{P_3}{1000} + \frac{2^2}{2} \right) = 700 \left(\frac{10 \times 10^3}{1000} + \frac{1.4^2}{2} \right) + 500 \left(\frac{5 \times 10^3}{1000} + \frac{2.5^2}{2} \right) - 1.5 \times 10^3$$

$$\frac{P_3}{1000} + \frac{2^2}{2} = 8.540416667$$

$$P_3 = 6540.416667$$

$$\approx 6.54 \text{ kPa}$$